TOR AMPLITUDE IN A BOUNDED INTERVAL IS EQUIVALENT TO REPRESENTABILITY BY A BOUNDED COMPLEX OF FLAT MODULES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof is written out in full: the first implication, the inductive truncation above b with the flatness of the kernel, and the truncation below a with the last-one-flat criterion.

Editorial note. The label lemma-tor-amplitude names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. This card belongs to the priority queue of the D-72 twin ruling (the sites-side statement is the designated counting side, now counted here for the first time). The statement and proof are copied from the pinned chapter file; the three cited inputs are kept as external pointers with their canonical Stacks tags.

Allowed standard prerequisites (editorial).

- Existence of bounded above complexes of flat modules. Same chapter, the section on flatness providing a bounded above complex of flat O-modules representing a given object. Supplies the representative K^bullet used in the second implication.
- Flatness of the kernel of a surjection of flat modules. Modules on Sites, the lemma stating that the kernel of a surjection between flat modules is flat. Supplies flatness of the kernel when truncating the representative.
- The flatness criterion for the last term of a bounded above complex. Same chapter, the lemma concluding flatness of the degree-a cokernel from the vanishing of the tensor cohomology below a. Supplies flatness of E^a after truncation from below.

Source reference. Original reference 0653: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 12403–12403. Prerequisite pointer invoked at proof line(s) 12467; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original section 00H9: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 2961–2961. Prerequisite pointer invoked at proof line(s) 12453; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

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Source reference. Original lemma 03EY: sources/stacks-chow/tags.txt, source lines 4473–4473. Prerequisite pointer invoked at proof line(s) 12458; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let $(\mathcal{C}, \mathcal{O})$ be a ringed site. Let E be an object of $D(\mathcal{O})$. Let $a, b \in \mathbf{Z}$ with $a \leq b$. The following are equivalent*

(1) *E has tor-amplitude in $[a, b]$.*
(2) *E is represented by a complex $\mathcal{E}^\bullet$ of flat $\mathcal{O}$-modules with $\mathcal{E}^i = 0$ for $i \notin [a, b]$.*

Proof. If (2) holds, then we may compute $E \otimes^{\mathbf{L}}_{\mathcal{O}} \mathcal{F} = \mathcal{E}^\bullet \otimes_{\mathcal{O}} \mathcal{F}$ and it is clear that (1) holds.

Assume that (1) holds. We may represent E by a bounded above complex of flat $\mathcal{O}$-modules $\mathcal{K}^\bullet$, see Section 00H9. Let n be the largest integer such that $\mathcal{K}^n \neq 0$. If $n > b$, then $\mathcal{K}^{n-1} \to \mathcal{K}^n$ is surjective as $H^n(\mathcal{K}^\bullet) = 0$. As $\mathcal{K}^n$ is flat we see that $\mathrm{Ker}(\mathcal{K}^{n-1} \to \mathcal{K}^n)$ is flat (Modules on Sites, Lemma 03EY). Hence we may replace $\mathcal{K}^\bullet$ by $\tau_{\leq n-1}\mathcal{K}^\bullet$. Thus, by induction on n, we reduce to the case that $K^\bullet$ is a complex of flat $\mathcal{O}$-modules with $\mathcal{K}^i = 0$ for $i > b$.

Set $\mathcal{E}^\bullet = \tau_{\geq a}\mathcal{K}^\bullet$. Everything is clear except that $\mathcal{E}^a$ is flat which follows immediately from Lemma 0653 and the definitions. $\square$